ORIGINAL MEMOIR · BOVINE TUBERCULOSIS

TRANSLATED FROM THE FRENCH

NEW EXPERIMENTAL RESEARCH ON THE VACCINATION OF CATTLE AGAINST TUBERCULOSIS

Thirty-four months of infecting cohabitation — 1912 to 1915

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Annales de l'Institut Pasteur, 34^e^ année, n° 9, septembre 1920, pp. 553–560

ABOUT THIS TRANSLATION

Translated in full from the printed French text of the Gallica digitisation of the Institut Pasteur copy (ark:/12148/bpt6k5791119x, views f1 to f8), working from the page images rather than the machine OCR, which corrupts the dates, doses and animal numbers on which the argument rests.

CONVENTIONS

Original page numbers are marked inline at each page break, and the original's asterisk dividers are kept. Paragraph breaks, heading wording, italics and running order follow the original. Within the text, translator's notes marked **TN** are the only matter that is not in the original; the display furniture is editorial — the subtitle, the running heads and folios, and the roman numerals on the section marks, the 1920 sections being unnumbered.

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In a previous memoir,[1] we showed that by culturing the bovine tubercle bacillus on glycerinated bile media, in long successive series, one obtains a race of attenuated bacilli, become *avirulent for the ox, the monkey and the guinea pig*, very well tolerated by the organism even at considerable doses injected into the veins, producing in no case tuberculous lesions, but nevertheless conferring on cattle, from the thirtieth day after vaccination, a durable resistance to challenge inoculations made by the intravenous route.

"It remains for us to establish," we said, "*the duration of this resistance in the face of natural contamination by continuous cohabitation with tuberculous cattle*. That is what we are studying at present."

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[1] These *Annales*, February 1913.

The experiments instituted for this purpose, begun on 21 November 1912, were brutally interrupted in August 1915 during the German occupation. We shall say later in what circumstances. Nevertheless, the results acquired up to that last date are interesting enough that we believe it useful to publish them.

We had at our disposal ten Breton heifers, aged nine to ten months, free of tuberculosis. Four were to serve us as controls. The six others were to be subjected to the vaccination trials.

On 21 November 1912, these six heifers received into the jugular vein a single vaccinal injection of 20 milligrammes of bovine bacilli (880 *millions of bacilli*) from a culture of the 70th passage on biled potato and two weeks old. That same day our heifers were placed, together with the four controls, in a byre specially arranged to favour natural contamination and already inhabited for two months by five adult tuberculous cows.

In order to ensure contamination, the tuberculous cows, subjected to a weekly change of place, were tied, as the figure below shows, immediately in front of the row of heifers under experiment. The sloping floor of the byre was arranged in such a way that the dejecta of the tuberculous cows, carried along by the urine, constantly soiled the litter and the feed set out in a common trough running the whole length of the row of heifers (see figure opposite).

The programme we had laid down further included a second vaccination (likewise of 20 *milligrammes* of biled bacilli) given at the end of one year's stay in the byre for numbers 41, 44, 47, and an identical third vaccination, carried out again after a further period of one year, for numbers 41 and 44.

The complement of five infecting cows was always maintained in full during the thirty-four months that the experi-

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ment lasted, which, on account of the deaths that occurred, made it necessary for ten cows presenting external signs of tuberculosis to pass through. Autopsy revealed in *eight* of them the existence of an advanced tuberculosis; in the two others only a discreet tuberculosis of the bronchial and mediastinal lymph nodes was found, perhaps excluding an effective contagion. One of these two animals remained in the byre for thirty-one months. It must therefore be accepted that only *three cows affected with severe tuberculosis were in the byre at any one time during the trials*.

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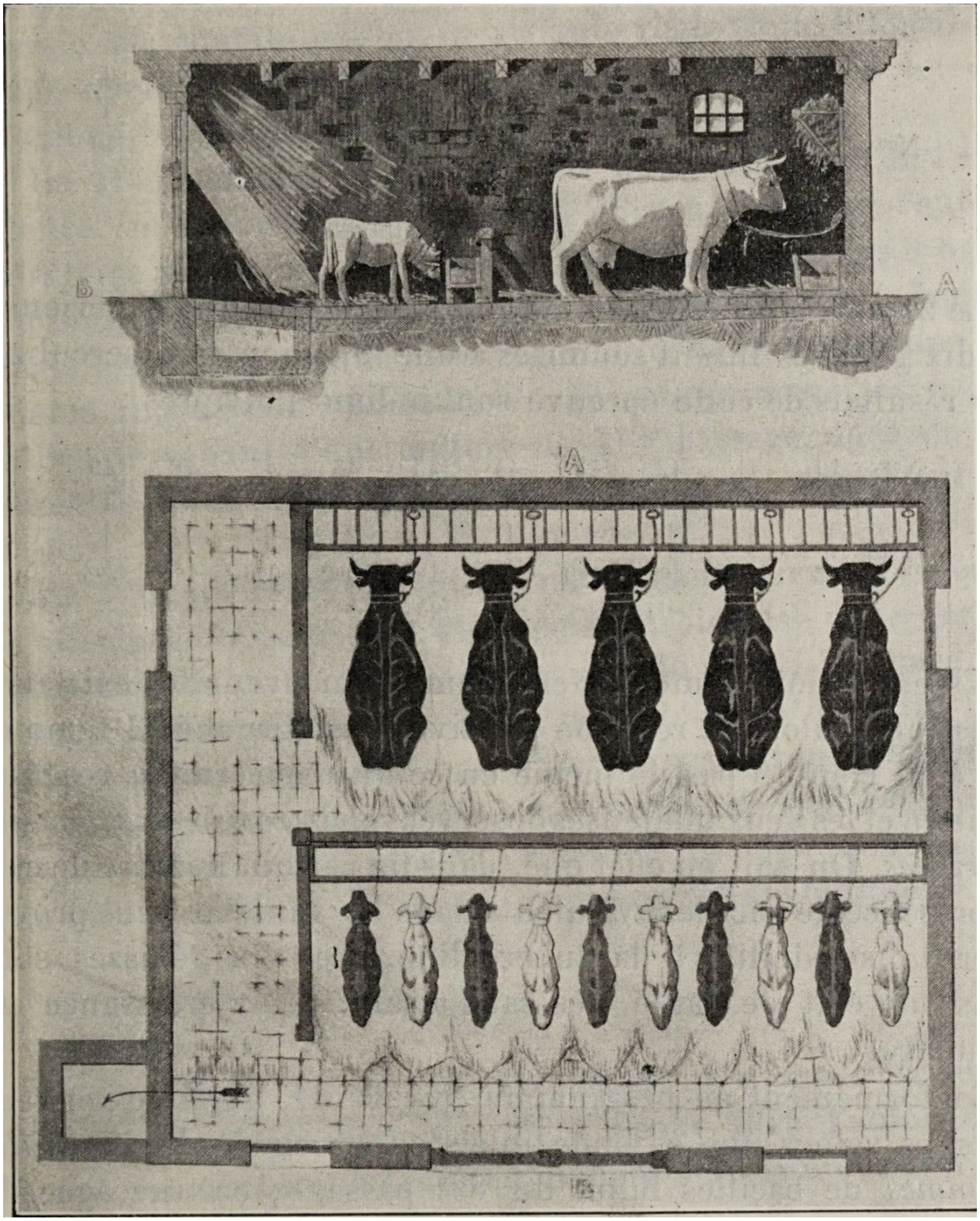
ARRANGEMENT OF THE ANIMALS IN THE CONTAMINATION BYRE

HEIFER NO.	40	41	42	43	44	45	46	47	48	49
LOWER ROW	T	V	V	T	V	V	T	V	V	T

Upper row, five stalls numbered 1 to 5: *infecting tuberculous cows*. Lower row, ten stalls as above. **T**, controls; **V**, vaccinated.

FIGURE, P. 555 — THE CONTAMINATION BYRE



An unnumbered plate, without printed caption, occupying most of p. 555. **Upper panel:** a cutaway perspective of the byre, an adult cow at the higher end (A) and a young heifer at the lower end (B), the sloping floor between them. **Lower panel:** a ground plan — an upper row of five animals, the infecting tuberculous cows (A), and below them a lower row of ten animals, the heifers under experiment (B), drawn alternately in solid and outline; the trough runs the length of the heifer row.

TN — The figure carries no caption in the original; the text of p. 554 refers to it as "the figure below" and "the figure opposite".

On 21 November 1913, one year after the start of the experiment, the ten heifers were subjected to an injection of tuberculin. The results of this test are given below:

TUBERCULIN TEST — 21 NOVEMBER 1913, TWELVE MONTHS

HEIFER NO.	40	41	42	43	44	45	46	47	48	49
GROUP	T	V	V	T	V	V	T	V	V	T
RISE	2°5	1°1	0°3	1°9	0°8	1°8	0°2	0°8	1°0	2°5
RESULT	+	–?	–	+	–	+	–	–	–?	+

TN — The figures are thermal rises in degrees, printed in the French convention (2°5 for 2.5 degrees); –? marks a doubtful result.

If it is permissible to interpret with confidence, in favour of the existence of tuberculosis, the positive reaction found in 3 controls out of 4, the same does not hold for the positive reaction and the two doubtful ones observed in *three* of the *six vaccinated heifers*. It is known, in fact, that in a certain number of cases the presence of avirulent bacilli in the organism can provoke sensitivity to tuberculin. The ten heifers are in a very satisfactory state of health. Their growth is regular.

In conformity with the programme, nos. 41, 44, 47 receive into the veins a second vaccinal injection of 20 *milligrammes* of biled bacilli of the 89th passage; culture 21 days old.

On 2 June 1914, eighteen months after the start of the experiment, the ten heifers are subjected to an injection of tuberculin. Here are the results of this test:

TUBERCULIN TEST — 2 JUNE 1914, EIGHTEEN MONTHS

HEIFER NO.	40	41	42	43	44	45	46	47	48	49
GROUP	T	V	V	T	V	V	T	V	V	T
RISE	1°4	0°7	0°6	2°5	0°5	0°9	0°5	0°7	0°2	1°2
RESULT	+	–	–	+	–	–	–	–	–	+

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After eighteen months of infecting cohabitation, the six vaccinated heifers, even those vaccinated only once at the very beginning, do not react to tuberculin, whereas three controls out of four are manifestly infected.

As for the fourth control, no. 46, which continues not to react, it must be accepted that individual resistance to contamination, non-existent in the face of experimental challenge inoculation, is nevertheless manifest where infecting cohabitation is concerned.

On 21 November 1914, two years after the start of the experiment, on account of the prohibition imposed on us by the German military authority against leaving our homes after 8 o'clock in the evening, the tuberculin injection could not be carried out; but heifers nos. 41 and 44 received into the veins a third vaccinal inoculation of 20 *milligrammes* of biled bacilli from a culture of the 113th passage, two weeks old. The health of all the animals remained perfect.

On 1 March 1915, following an accident to the tethering collar of heifer no. 41 (vaccinated three times), the latter was found strangled at two o'clock in the afternoon. The autopsy, made immediately, did not allow the least tuberculous lesion to be detected. The bronchial lymph nodes were taken in their entirety, ground up and inoculated under the skin of eight guinea pigs. On 23 July 1915 the guinea pigs show no local lesions; they are sacrificed and found free of tuberculosis.

* * *

The experiment was proceeding under good conditions when, in the first days of August 1915, the German authority had an order posted requiring the declaration, under the severest penalties, of all cattle existing on the territory of the commune, these being liable to requisition by the army.

To avoid this declaration, we decided to bring the experiment to an end and to proceed to the clandestine slaughter of our animals.

On 16 August 1915, thirty-two months after the start of the trials,

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the nine remaining heifers were subjected to an injection of tuberculin:

TUBERCULIN TEST — 16 AUGUST 1915, THIRTY-TWO MONTHS

HEIFER NO.	40	41	42	43	44	45	46	47	48	49
GROUP	T	V	V	T	V	V	T	V	V	T
RISE	2°9	*dead*	2°5	2°6	0°0	1°7	0°3	0°7	0°8	1°7
RESULT	+	*free of disease*	+	+	–	+	–	–	–	+

TN — No. 41 is the heifer strangled on 1 March 1915; her column records the death and the negative autopsy ("free of disease") rather than a tuberculin result.

It can be concluded from this test that the heifers vaccinated *three times* in 1912, 1913 and 1914 (nos. 41 and 44), and the one vaccinated *twice* in 1912 and 1913 (no. 47), are still free of tuberculosis; but that, of the three other heifers (nos. 42, 45 and 48) vaccinated *only once* in 1912, at the start of the experiment, two out of three (nos. 42 and 45) are contaminated, and that they became so between the eighteenth and the thirty-second month of infecting cohabitation.

As for the fourth control, no. 46, it still does not react, thus showing an example of individual resistance that is altogether remarkable.

From 27 August to 1 October 1915, the slaughter of all the animals was carried out clandestinely, beginning with the controls and with those which had reacted to the last tuberculin test.

I A. — CONTROLS

No. 40. — Very discreet lesions. In a mesenteric lymph node, one caseous tubercle the size of a millet grain. In the bronchial lymph nodes, 2 caseous tubercles the size of a millet grain.

No. 43. — In the right lung, 9 tubercles ranging from the size of a hazelnut to that of a walnut, all caseous. In the left lung, 7 tubercles the size of a small hazelnut. In the bronchial lymph nodes, 7 tubercles the size of hempseed grains.

No. 49. — The mediastinal lymph nodes are tripled in volume and show on section numerous tuberculous foci ranging from the size of a millet grain to that of a hazelnut. The bronchial lymph nodes show on section several tubercles the size of hempseed grains.

No. 46. (Never reacted to the tuberculin test.) — Entirely free of tuberculosis. The bronchial lymph nodes are taken, ground up and inoculated under the skin of 4 guinea pigs which, on 2 December 1915, are sacrificed and show no lesion.

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II B. — VACCINATED

SERIES I — VACCINATED ONLY ONCE, IN 1912

No. 42. — In the 2 lungs are found 15 tubercles ranging from the size of a pea to that of a hazelnut, all caseous. The bronchial and mediastinal lymph nodes are stuffed with lesions.

No. 45. — In a mesenteric lymph node, a tuberculous lesion the size of a hazelnut. In a mediastinal lymph node, a focus the size of a small pea. Bronchial lymph nodes healthy in appearance. In the right lung, 2 tubercles the size of a hazelnut.

No. 48. — No apparent tuberculous lesion. The bronchial lymph nodes are taken, ground up and inoculated under the skin of 5 guinea pigs which, subsequently, show no tuberculous lesions.

SERIES II — VACCINATED TWICE, IN 1912 AND 1913

No. 47. — No apparent tuberculous lesion. The bronchial lymph nodes are taken, ground up and inoculated under the skin of 5 guinea pigs. On 2 December 1915 these animals, sacrificed, show no lesion.

SERIES III — VACCINATED THREE TIMES, IN 1912, 1913 AND 1914

No. 41. — Died accidentally on 1 March 1915. The autopsy and the inoculation of the bronchial lymph nodes showed that this animal was free of tuberculosis.

No. 44. — No apparent tuberculous lesion. The bronchial lymph nodes are taken, ground up and inoculated under the skin of 6 guinea pigs. On 2 December 1915 these guinea pigs, sacrificed, show no tuberculous lesion.

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III CONCLUSIONS

1. The culture of the bovine tubercle bacillus in successive series on glycerinated bile media (we have maintained these cultures regularly for 12 and a half years) makes it possible to obtain a race of *non-tuberculigenic* bacilli, perfectly tolerated by the organism of cattle and by that of other animals susceptible to the tuberculous virus.

2. This avirulent race behaves as a true vaccine, in the sense that *inoculated at a suitable dose into the veins of cattle, it confers on these animals a tolerance which is manifest*, not only in the face of experimental challenge inoculation, but also *with regard to contamination by close cohabitation in infected byres*.

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3. This tolerance, linked, we believe, to the presence of the avirulent bacilli in the organism, does not exceed *18 months* after a single vaccination, but *it can be maintained by revaccinations carried out each year*, which are, in themselves, *harmless*.

* * *

By other experiments, which do not fall within the scope of this work, we have acquired the certainty that our living biled bovine bacillus is *harmless to man*, even by intravenous inoculation at a dose of at least 44,000 bacilli.

Under these conditions it can be affirmed that if vaccination by means of living tubercle bacilli, rendered non-tuberculigenic by serial culture on bile, proves effective after a sufficiently prolonged experimentation, it will present none of the drawbacks which caused the abandonment, with good reason, of the procedure of *Behring* (bovovaccination) and that of *Robert Koch* and *Schultze* (tauruman), which are based on the use of human or bovine bacilli virulent for man and for the ox.

It would therefore be desirable that the trial of this method could be extended to a greater number of animals and pursued over a cycle of years corresponding to the average duration of the life of cattle, in order to establish its practical value.

Unfortunately the present economic conditions oblige us to await circumstances more favourable to the purchase and the maintenance of a sufficiently large herd.

AUTHORS

A. Calmette & C. Guérin. The 1920 byline carries no affiliation; the Institut Pasteur de Lille, printed on the 1924 memoir, is not stated here.

TRANSLATION

Complete English translation of the memoir published in the Annales de l'Institut Pasteur, 34^{e} année, n° 9, September 1920, pp. 553–560. The article immediately following in the same issue, at p. 561, is Jules Bordet on the theories of blood coagulation, published for the Metchnikoff jubilee.